Wi-Fi and Bluetooth Technology

Introduction

Wi-Fi and Bluetooth are the two most common wireless technologies that connect our devices without cables. Both use radio waves and are found in smartphones, laptops, TVs, watches, earbuds, cars and smart home devices. They serve different purposes: Wi-Fi connects devices to a network and the internet over a larger area with high speeds, while Bluetooth connects devices directly to each other over short distances with low power. Both belong to the family of wireless technologies that also includes cellular networks such as 4G and 5G, and NFC.

What Is Wi-Fi?

Wi-Fi is a wireless networking technology that allows devices to connect to a local area network (WLAN), and through a router to the internet, using radio waves. It is based on the IEEE 802.11 family of standards. The Wi-Fi Alliance, formed in 1999, certifies products and owns the Wi-Fi trademark. The name "Wi-Fi" was created by a branding agency and is often wrongly thought to mean "wireless fidelity"; it does not stand for anything. A Wi-Fi network consists of a router or access point, connected to the internet by broadband such as fibre, which broadcasts a signal identified by a network name (SSID). Devices connect with a password, and data is encrypted using security protocols such as WPA2 and WPA3. A typical home router covers about 30 to 50 metres indoors, but walls, floors and other devices reduce the range. Mesh Wi-Fi systems use several units to cover large homes.

What Frequency Bands Does Wi-Fi Use?

Wi-Fi uses three main frequency bands. The 2.4 GHz band has a longer range and better wall penetration but offers lower speeds and is crowded, because many devices such as Bluetooth devices, microwave ovens and cordless phones also use it. The 5 GHz band offers higher speeds and less interference but a shorter range. The 6 GHz band, which was introduced with Wi-Fi 6E and is also used by Wi-Fi 7, provides even more channels and very high speeds with very little congestion, but it has a shorter range and needs newer devices. India has been opening up part of the 6 GHz band for Wi-Fi use. Each band is divided into channels, and routers pick the best channel. Dual-band and tri-band routers support more than one band.

Wi-Fi Generations: What Is the Difference Between Wi-Fi 5 and Wi-Fi 6?

Wi-Fi standards have been given simple generation names. Wi-Fi 4 is 802.11n (2009), Wi-Fi 5 is 802.11ac (2014), Wi-Fi 6 is 802.11ax (2019), Wi-Fi 6E extends Wi-Fi 6 to the 6 GHz band, and Wi-Fi 7 is 802.11be (2024).

Wi-Fi 5 (802.11ac) works only in the 5 GHz band, with a maximum theoretical speed of about 3.5 Gbps, and uses MU-MIMO for downloads only and 256-QAM modulation. Wi-Fi 6 (802.11ax) works in both 2.4 GHz and 5 GHz, with a maximum theoretical speed of about 9.6 Gbps, which is about 40 percent faster than Wi-Fi 5 in raw speed, but the main improvement is efficiency in crowded places, such as homes with many devices, offices, stadiums and campuses. It introduces OFDMA, which splits a channel into smaller sub-channels so that multiple devices can be served at the same time, MU-MIMO for both uploads and downloads, 1024-QAM for higher data density, Target Wake Time (TWT), which lets devices sleep and save battery, and better WPA3 security. Compared to Wi-Fi 5, Wi-Fi 6 gives better speeds, less lag with many devices, longer battery life for devices and better performance at the edge of coverage. Wi-Fi 7 adds 320 MHz channels, 4096-QAM and multi-link operation, with a theoretical speed of about 46 Gbps. All Wi-Fi versions are backward compatible, but to get the benefits you need both a router and devices that support the new version.

What Is Bluetooth?

Bluetooth is a short-range wireless technology for exchanging data between devices over short distances, using radio waves in the 2.4 GHz ISM band. It was invented in 1994 by engineers at Ericsson, including Jaap Haartsen, as a wireless replacement for cables, and the Bluetooth Special Interest Group (SIG) was formed in 1998. The name comes from Harald "Bluetooth" Gormsson, a 10th-century Danish king who united Denmark and Norway, symbolising the uniting of different devices, and the Bluetooth logo combines the Nordic runes for his initials H and B. Devices connect by a process called pairing and form small networks called piconets. Bluetooth uses frequency hopping, changing channels about 1,600 times a second, to avoid interference. Common uses are wireless earbuds and headphones, speakers, smartwatches and fitness bands, keyboards and mice, car hands-free systems, file sharing, and connecting smart home devices, medical devices and location trackers.

What Is the Range of Bluetooth?

The range of Bluetooth depends on the power class and version. A typical Class 2 device, such as a phone or earbuds, has a range of about 10 metres (about 33 feet). Class 1 devices, used in industrial products, can reach up to about 100 metres, and Class 3 devices only about 1 metre. Bluetooth 5.0 and later versions, which use Bluetooth Low Energy (BLE), can reach up to about 240 metres in open space under ideal conditions, with higher speed and larger capacity, though the real-world range in homes is much shorter because of walls and interference. Bluetooth Classic is used for audio streaming, and BLE uses very little power, and is used for wearables and sensors. Newer versions have LE Audio, which allows broadcast audio, with the LC3 codec, and Auracast, allowing many people to listen to the same audio source. Bluetooth data rates range from about 1 to 3 Mbps for classic Bluetooth and up to 2 Mbps in BLE, much lower than Wi-Fi.

Wi-Fi Versus Bluetooth

Wi-Fi is designed for high-speed internet and network access over a range of tens of metres, with higher power consumption, and typically connects devices through a router. Bluetooth is designed for short-range, device-to-device links with low power consumption and lower speeds. They both use 2.4 GHz, so they can interfere, but modern devices manage this. Wi-Fi Direct and Bluetooth tethering allow other connections, for example sharing a mobile hotspot.

Security

To be safe, use WPA3 or WPA2 with a strong password on the home router, change the default router password, keep the firmware updated, avoid sensitive transactions on open public Wi-Fi without a VPN, turn off Bluetooth and Wi-Fi when not needed, don't accept unknown pairing requests, and be careful about fake hotspots with names like those of cafes.

Wi-Fi and Bluetooth in India

India has a large number of broadband and Wi-Fi users, with fibre broadband spreading. The government launched PM-WANI (Prime Minister's Wi-Fi Access Network Interface) in December 2020 to enable small shops and entrepreneurs to offer public Wi-Fi hotspots, and the Chennai Metro and many public places offer free Wi-Fi. India is a very large market for Bluetooth earbuds and wearables, with Indian brands such as boAt, Noise and Zebronics selling millions of units.

Challenges and Future

Challenges include interference, security and range limits. The future includes Wi-Fi 7 and beyond, indoor positioning with Bluetooth, Wi-Fi sensing, and the combination of Wi-Fi with 5G.

Quick Facts

- Wi-Fi uses the 2.4 GHz, 5 GHz and 6 GHz bands.
- A typical Bluetooth range is about 10 metres; Class 1 reaches about 100 metres and Bluetooth 5 can reach about 240 metres in open space.
- Wi-Fi 6 improves on Wi-Fi 5 with OFDMA, MU-MIMO in both directions, TWT and better performance with many devices.
- Bluetooth is named after King Harald Bluetooth of Denmark.
- India launched PM-WANI for public Wi-Fi in December 2020.